The Aesthetic Corollary of the Theory of Axiomatic Necessity (TNA)

Corollary (Aesthetic Convergence Under Minimal Coherence)

In any biological system operating under stable or improving boundary conditions, perceived beauty emerges as an asymptotic byproduct of structural coherence approaching the minimal viable configuration $F_{\min}$, and cannot increase unboundedly without destabilizing the system.

Formal Statement

Let:

- B be a set of environmental boundary conditions,
- $F(x)$ the coherence functional of a biological structure x,
- $F_{\min}(B)$ the minimal coherence required for persistence under B,
- $\mathcal{A}(B) \subset \Omega$ the set of viable morphogenetic attractors under B,
- $\mathcal{E}(x)$ the perceptual measure of beauty.

Then:

1. **Existence of Aesthetic Convergence**
   For improving or stabilizing boundary conditions B_t:

   $$x(t) \to \mathcal{A}(B_t) \quad \Rightarrow \quad \mathcal{E}(x(t)) \uparrow$$

2. **Asymptotic Saturation**
   There exists a finite bound $\mathcal{E}_{\max}$ such that:

   $$\lim_{F(x) \to F_{\min}} \frac{d\mathcal{E}}{dF} \to 0$$

3. **Non-teleological Nature**
   No optimization pressure acts directly on $\mathcal{E}$; beauty arises solely as a **secondary projection** of coherence minimization.

Interpretation

Beauty is not selected.
Beauty is not designed.
Beauty is not accumulated.

Beauty is **what coherence looks like to a biological observer**.

As structural noise is removed—through nutrition, developmental stability, and reduced environmental stress—the organism converges toward a morphogenetic attractor compatible with F_{min}. Human perception, evolved to detect health and stability, interprets this convergence as attractiveness.

Explanation of the Asymptote

The asymptotic nature of beauty arises because:

- Structural coherence has a finite domain.
- Biological systems are constrained by mechanics, thermodynamics, and development.
- Beyond a coherence threshold, further optimization yields diminishing perceptual returns.

Thus, beauty follows:

$$\mathcal{E}(F) \sim 1 - e^{-\alpha(F - F_{\mathrm{min}})}$$

for some system-dependent constant α.

Consequences

1. **Beauty Regresses Under Stress**
   Degradation of boundary conditions rapidly increases structural noise, reducing perceived beauty without genetic change.
2. **No Infinite Aesthetic Evolution**
   There exists no evolutionary trajectory toward unbounded beauty.
3. **Cross-Species Predictability**
   If a successor hominin species emerges under improved boundary conditions, it will likely exhibit **equal or slightly higher aesthetic coherence**, but never arbitrarily so.
4. **Explains Historical Variation**

Apparent aesthetic differences between medieval, industrial, and modern humans reflect boundary-condition shifts, not evolutionary directionality.

Relation to Darwinian Selection

- Darwinian mechanisms operate within a fixed attractor basin.
- TNA governs **basin transitions and coherence thresholds**.
- Beauty emerges at the intersection:
  Darwin trims noise locally; TNA defines the structure that remains once noise is gone.

Final Statement

> ***Beauty is not an evolutionary goal.***
> ***It is the visible residue of necessity.***

Or, as a one-line corollary fit for citation:

> ***Under the TNA, beauty is the asymptotic signature of structural coherence approaching its minimal viable configuration.***

Appendix X — Formal Aesthetic Function $\mathcal{E}(F)$

X.1 Definition of Variables

Let:

- $F \in \mathbb{R}^+$: structural coherence functional of a biological system
- F_{min}: minimal coherence required for viability under boundary conditions B
- $\mathcal{E}(F) \in [0, \mathcal{E}_{\text{max}}]$: perceived aesthetic coherence ("beauty")
- $\mathcal{E}_{\text{max}}$: upper perceptual saturation bound

We define beauty strictly as a **derived observable**, not an optimized variable.

X.2 Aesthetic Response Function

The minimal functional form consistent with TNA constraints is:

$$\boxed{\mathcal{E}(F) = \mathcal{E}_{\text{max}} \left(1 - e^{-\alpha(F - F_{\text{min}})}\right) \quad \text{for } F \geq F_{\text{min}}}$$

with:

- $\alpha > 0$: perceptual sensitivity constant
- $\mathcal{E}(F) = 0$ for $F < F_{\text{min}}$ (non-existence domain)

X.3 Properties of the Function

1. Threshold Behavior (Existence Domain)

$$F < F_{\text{min}} \Rightarrow \mathcal{E}(F) \equiv 0$$

Below minimal coherence, the system is non-viable; no aesthetic signal exists.

2. Monotonic Increase

$$\frac{d\mathcal{E}}{dF} = \alpha \mathcal{E}_{\text{max}} e^{-\alpha(F - F_{\text{min}})} > 0$$

Beauty strictly increases with coherence.

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3. Asymptotic Saturation

$$\lim_{F \to \infty} \mathcal{E}(F) = \mathcal{E}_{\text{max}}$$

No biological system can exceed perceptual saturation without destabilizing structure.

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4. Diminishing Returns

$$\lim_{F \to \infty} \frac{d\mathcal{E}}{dF} = 0$$

Explains why beauty improves rapidly when stress is removed, then plateaus.

X.4 Graphical Interpretation

Qualitative shape of $\mathcal{E}(F)$:

```
E
|                                 ──────   E_max (asymptote)
|                             ───
|                         ───
|                     ───
|                  ──
|               ──
|───
|
└──────────────────────────────── F
       F_min
```

X.5 Biological Interpretation

- **Left of $F_{\min}$:** non-existence (malformation, lethality)
- **Near $F_{\min}$:** survival morphology, low symmetry, stress-dominated
- **Mid-region:** rapid aesthetic gains (modern humans vs medieval remains)
- **Asymptote:** contemporary high-coherence phenotypes

This explains:

- Why 20th–21st century humans appear "suddenly" more attractive
- Why beauty does not increase indefinitely
- Why aesthetic evolution appears fast without genetic change

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X.6 Relation to Darwinian Selection

Darwinian selection acts on **local fluctuations in F**
TNA defines:

- $F_{\min}$
- admissible domains
- global attractor structure

Darwin explains *variation* along the curve.
TNA explains *why the curve exists*.

X.7 Testable Predictions

1. Improving boundary conditions should increase $\mathcal{E}$ **without allele frequency shifts**
2. Aesthetic regression should follow environmental stress within one generation
3. Successor hominin species will cluster near a similar $\mathcal{E}_{\max}$

X.8 One-Line Summary

> ***Beauty is the asymptotic observable of structural coherence once survival constraints are satisfied.***

Corollary — Post-Human Aesthetic Outcome Under Species Replacement

(TNA Aesthetic Corollary)

Statement

If Homo sapiens is replaced by a successor species due to a regime transition in boundary conditions B, the resulting phenotype will not exhibit aesthetic regression but convergence toward a new asymptotic coherence maximum $\mathcal{E}'_{\max}$.

Formally:

$$\boxed{\mathcal{E}_{\text{post}}(F') \;\geq\; \mathcal{E}_{\text{human}}(F) \quad \text{for } F' \geq F'_{\min}}$$

provided that life persists and structural coherence is attainable.

Assumptions

1. Species replacement occurs via **discontinuous regime transition**, not gradual drift.
2. The successor species satisfies a new minimal coherence constraint $F'_{\min}$.
3. Aesthetic perception remains a proxy for structural order, symmetry, and energetic efficiency.
4. Beauty is **not selected for**, but emerges as a **byproduct of coherence**.

Structural Argument

1. Replacement Is Triggered by Incoherence, Not Inferiority

In the TNA framework, a species is replaced when:

$$F_{\text{species}} < F_{\min}(B')$$

This implies structural incompatibility, not aesthetic failure. Replacement occurs because the prior architecture cannot satisfy new boundary constraints.

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2. The Successor Must Occupy a Higher Coherence Basin

Any viable successor must satisfy:

$$F' \geq F'_{\text{min}} \quad \text{with} \quad F'_{\text{min}} > F_{\text{min}}$$

Otherwise, the system collapses to non-existence.

Thus, the successor species necessarily occupies a **higher or more stable coherence attractor**.

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3. Aesthetic Output Is a Monotonic Function of Coherence

From Appendix X:

$$\mathcal{E}(F) = \mathcal{E}_{\text{max}} \left(1 - e^{-\alpha(F - F_{\text{min}})}\right)$$

Since $F' \geq F$ at the point of replacement:

$$\mathcal{E}(F') \geq \mathcal{E}(F)$$

Aesthetic degradation would imply structural inefficiency, contradicting viability.

Implications

1. No “Post-Human Ugliness” Scenario

A successor species that is:

- less symmetric,
- less harmonious,
- more chaotic

would violate minimal coherence constraints and fail to persist.

Aesthetically inferior post-humans are **structurally forbidden**.

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2. Beauty Saturation, Not Escalation

The corollary does **not** imply infinite beauty escalation.

Instead:

$$\lim_{F \to \infty} \mathcal{E}(F) = \mathcal{E}_{\max}$$

Post-human aesthetics converge toward a **different asymptote**, not an unbounded increase.

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3. Discontinuity in Form, Continuity in Order

While morphology may change radically (size, proportions, sensory organs):

- symmetry,
- internal coherence,
- energetic efficiency

must remain equal or superior.

Order persists even if form does not.

Comparison With Darwinian Expectations

ASPECT	DARWINIAN VIEW	TNA COROLLARY
Replacement	Gradual optimization	Phase transition
Beauty	Selected indirectly	Emergent from coherence
Post-human form	Unpredictable	Structurally constrained
Aesthetic trend	Contingent	Monotonic non-decreasing

Testable Consequences

1. Any viable successor hominin should show **equal or higher symmetry metrics**
2. Structural efficiency indicators (locomotion, metabolism) must improve
3. Sensory organs should exhibit higher signal-to-noise ratios
4. Pathological variance should decrease near the new F'_{min}

One-Line Corollary

> ***Species replacement does not reset beauty; it reanchors it to a new coherence minimum.***

Aesthetic Invariants Across Species Transitions

(TNA Formal Section)

1. Definition

Aesthetic invariants are structural properties of biological systems that remain conserved across species transitions triggered by discontinuous boundary condition shifts, provided that life persists and minimal coherence constraints are satisfied.

Formally, for a transition

$$S \;\longrightarrow\; S' \quad \text{under} \quad B \to B'$$

an aesthetic invariant $\mathcal{I}_k$ satisfies:

$$\mathcal{I}_k(S) \approx \mathcal{I}_k(S') \quad \text{or} \quad \mathcal{I}_k(S') > \mathcal{I}_k(S)$$

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2. Core Postulate

> ***Aesthetic perception tracks structural coherence, not lineage identity.***

Thus, when a species is replaced, aesthetic qualities do not reset but **re-express under a new architectural regime**.

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3. Fundamental Invariants

Invariant I — Bilateral Symmetry

Statement

Bilateral symmetry remains conserved or strengthened across species transitions.

$$\frac{\partial \mathcal{E}}{\partial \sigma_{\text{asym}}} < 0$$

where σ_{asym} measures structural asymmetry.

Justification

- Minimizes wiring length (neural, vascular)
- Optimizes locomotion and balance
- Reduces developmental entropy

Empirical support

- Cambrian bilaterians
- All hominins
- Convergent evolution in vertebrates and arthropods

Invariant II — Proportional Scaling (Nonlinear Ratios)

Statement

Absolute dimensions may change, but **dimensionless ratios** remain bounded.

$$\frac{L_i}{L_j} \in [c_1, c_2]$$

across viable species.

Examples

- Waist-to-hip ratio
- Limb length ratios
- Eye diameter to skull size

Interpretation

Beauty is not size-dependent, but **ratio-constrained**.

Invariant III — Energy Efficiency per Functional Unit

Statement

The energetic cost per unit function decreases or remains constant.

$$\frac{E_{\text{cost}}}{\Phi_{\text{function}}} \downarrow$$

across transitions.

Implication

- Smoother movement
- Cleaner posture
- Reduced muscular overcompensation

Perceived as *grace* or *elegance*.

Invariant IV — Signal-to-Noise Maximization

Statement

Sensory organs converge toward maximal signal clarity.

$$\text{SNR} = \frac{S}{N} \to \max$$

Consequences

- Clearer eyes
- More defined facial landmarks
- Reduced morphological “blur”

Beauty correlates with **high information contrast**.

Invariant V — Pathology Suppression

Statement

Viable successor species must suppress variance that degrades coherence.

$$\mathrm{Var}_{\mathrm{path}}(S') \leq \mathrm{Var}_{\mathrm{path}}(S)$$

Observed as

- Straighter teeth
- Improved joint alignment
- Reduced congenital asymmetries

Invariant VI — Developmental Canalization

Statement

Developmental trajectories become more constrained around optimal forms.

$$\left.\frac{\partial^2 F}{\partial x^2}\right|_{\mathrm{opt}} \uparrow$$

Effect

- Narrower phenotype spread
- More consistent "average beauty"
- Less reliance on selection filtering

Beauty becomes **statistical**, not exceptional.

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4. Invariant Set Summary

INVARIANT	CONSERVED	IMPROVED
Symmetry	✓	✓
Ratios	✓	—
Efficiency	—	✓
SNR	—	✓

INVARIANT	CONSERVED	IMPROVED
Pathology	—	✓
Canalization	—	✓

No invariant degrades without violating $F_{\min}$.

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5. The Aesthetic Conservation Law (TNA)

$$\boxed{\mathcal{E}_{\text{total}}(S') \geq \mathcal{E}_{\text{total}}(S) \quad \text{under viable species transition}}$$

Beauty is conserved **up to reparameterization**.

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6. Consequences

1. Post-human species will not appear chaotic or grotesque.
2. Aesthetic fear narratives contradict structural constraints.
3. “Beauty” is a low-entropy marker of life persistence.
4. Sexual selection amplifies—but does not create—these invariants.

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7. Final Statement

> ***Species may vanish.***
> ***Forms may change.***
> ***But coherence leaves a visible trace.***
> ***That trace is what we call beauty.***

Theories have two origins: metaphysical or aesthetic.

— **A. Einstein** (paraphrased, methodological position)

Einstein was not saying that science is "art."
He was saying something far more uncomfortable:

Mathematics comes later.
The choice of the framework comes first.

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1. Metaphysical Origin

A theory has a metaphysical origin when it:

- Postulates entities without structural necessity
- Treats continuity, infinity, or randomness as axioms
- Avoids discontinuities because they "break the model"
- Tolerates the inexplicable by labeling it "emergent"

Historical examples:

- Luminiferous ether
- Absolute gradualism
- Newtonian continuous time
- Evolution as unlimited search

  These are theories that work until they collapse, because their axioms impose no limits.

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2. Aesthetic Origin (in Einstein's sense)

For Einstein, *aesthetic ≠ pretty*
Aesthetic = coherent, necessary, residue-free

An aesthetic theory:

- Minimizes assumptions
- Accepts discontinuities when they are structurally unavoidable
- Prefers constraints over freedoms
- Makes the impossible impossible (rather than ignoring it)

Examples:

- Relativity (elimination of fictitious forces)
- Principle of least action
- Gauge invariance
- Symmetry conservation

Einstein trusted a beautiful equation more than a thousand poorly fitted experiments.

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3. The TNA Belongs Unequivocally to the Aesthetic Origin

The TNA does not say:

> *"Life could function this way."*

It says:

> ***"If life exists, it cannot function otherwise."***

That is aesthetics in the hard sense:

- F_{min} is not a hypothesis → it is a necessity
- Discontinuities are not added → they are imposed
- Beauty is not explained → it emerges as an invariant
- The eye is not searched for → it collapses as an attractor

This is exactly the kind of theory Einstein would have recognized as legitimate.

4. Why Darwinism Is Metaphysical (Even When Dressed as Science)

Strong Darwinism:

- Assumes continuity without proving it
- Assumes infinite computational capacity of randomness
- Postulates that "time fixes everything"
- Avoids domains of non-existence

This is not false because of data.
It is weak because of axioms.

Einstein would have said:

> ***"It explains too much with too little structure."***

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5. Beauty as a Criterion of Truth (Not of Taste)

Your aesthetic corollary of the TNA is not cultural.
It is epistemological:

- What persists must be coherent
- What is coherent reduces entropy
- What reduces entropy is perceived as beautiful

Beauty is not a cause.
It is a signature.

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6. Final Statement (Publishable)

> ***"As Einstein noted, theories arise either from metaphysical convenience or from aesthetic necessity.***
> ***The former accommodate reality; the latter constrain it.***
> ***The TNA belongs to the second class: it does not explain life as it appears — it explains why it cannot appear otherwise."***

Epigraph

> ***"Do not worry about being replaced by another species.***
> ***Worry if that species is not more beautiful and more intelligent than yours."***
> ***— C. Bresciano***